

2025 HKDSE Physics

Paper 1A — Multiple-choice Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Question 1

There are 33 questions. Questions marked with * involve knowledge of the extension component. Take $g = 9.81 \text{ m s}^{-2}$.

An equal amount of energy is supplied to each of the substances P, Q, R and S of mass and specific heat capacity listed below. Which substance would have the largest rise in temperature?

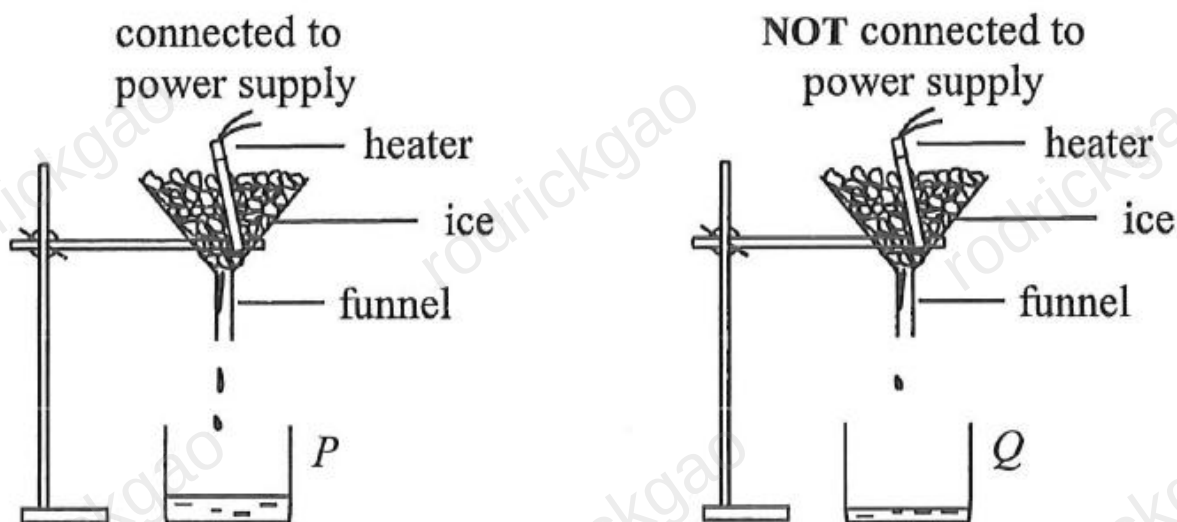
Option	Substance	Mass / kg	Specific heat capacity / $\text{J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$
A.	P	2	4200
B.	Q	3	2100
C.	R	4	900
D.	S	5	840

Official answer: C

For the same heat input, $\Delta T = Q/(mc)$. R has the smallest heat capacity: $mc = 4 \times 900 = 3600 \text{ J }^{\circ}\text{C}^{-1}$; hence its temperature rise is greatest. Answer: C.

Question 2

The set-up below can be used to find the specific latent heat of fusion of ice, l_f . The energy supplied by the heater is measured by a joulemeter. The mass of ice melted by the heater is given by the difference between the mass of water collected in beakers P and Q.



Question 2 set-up

How would the calculated value of l_f be affected if

(I) the heater is not completely covered by the ice?

(II) the mass of water collected in beaker P is taken as the mass of ice melted by the heater?

A. (I) decrease; (II) increase

B. (I) decrease; (II) decrease

C. (I) increase; (II) increase

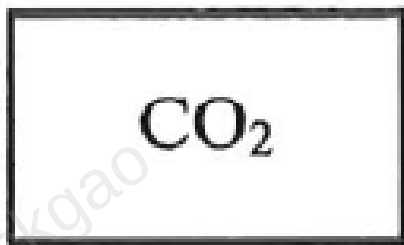
D. (I) increase; (II) decrease

Official answer: D

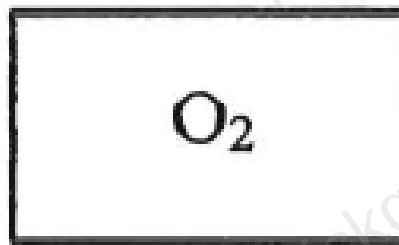
$l_f = E/m$. With the heater partly exposed, energy lost to the air is included in E , so l_f is overestimated. Using mass P includes ice melted naturally, so m is too large and l_f is underestimated. Answer: D.

Question 3

Identical closed vessels X and Y contain carbon dioxide (CO_2) and oxygen (O_2) respectively.



vessel X



vessel Y

Question 3 vessels

Both gases are at room temperature and atmospheric pressure. Which of the following physical quantities of the gas molecules in the two vessels is/are the same?

- (1) root-mean-square speed
- (2) average kinetic energy
- (3) frequency of collision with the walls of the vessel

A. (1) only

B. (2) only

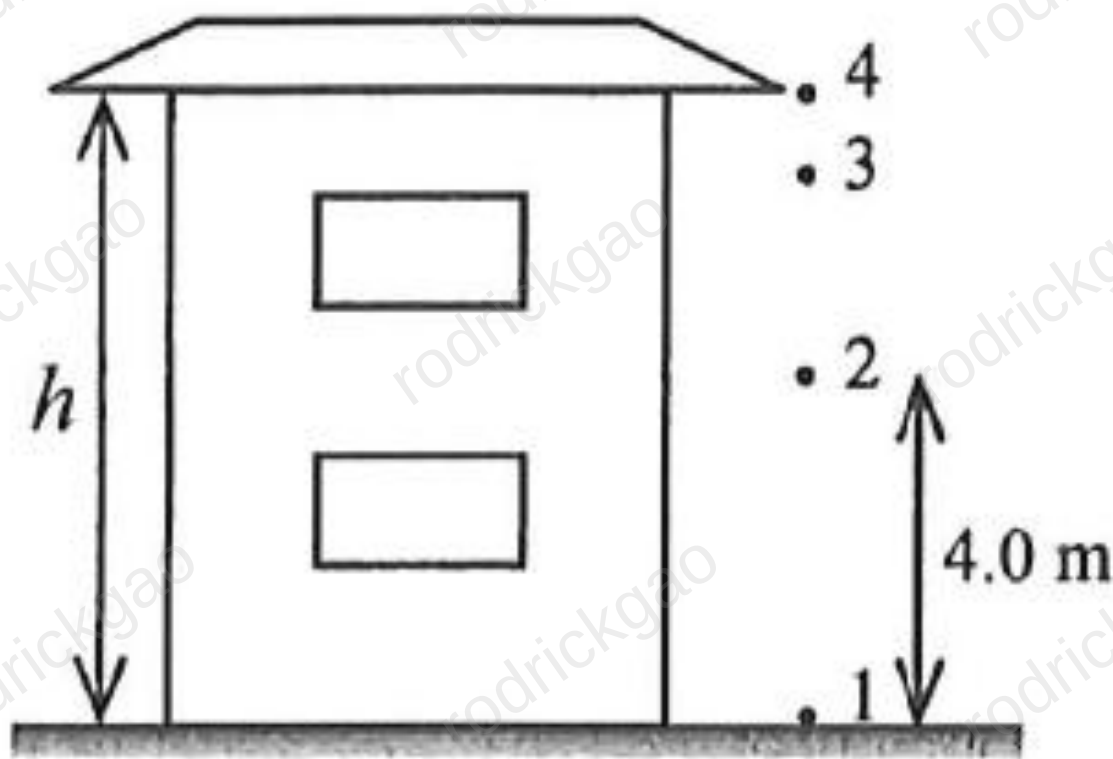
C. (1) and (3) only

D. (2) and (3) only

Official answer: B

At the same temperature, the mean molecular kinetic energy is the same. Since $v_{\text{rms}} \propto 1/\sqrt{M}$, the collision frequency also differs. Only (2) is correct. Answer: B.

Question 4



Question 4 falling droplets

Water drips regularly at a constant rate from a building's roof, which is at a height h above the ground. The droplets fall from rest such that the first droplet reaches the ground just when the fourth droplet leaves the roof as shown. If the second droplet is 4.0 m above the ground at this instant, find h . Neglect air resistance.

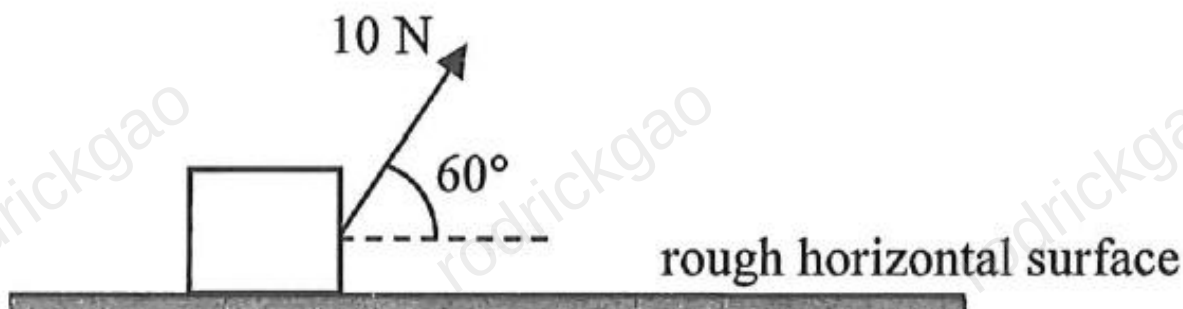
- A. 5.8 m
- B. 6.4 m
- C. 7.2 m
- D. 7.8 m

Official answer: C

Let the drop interval be τ . The first and second drops fall for 3τ and 2τ respectively. Thus $h - 4 = \frac{1}{2}g(2\tau)^2 = (4/9)h$, giving $h = 7.2$ m. Answer: C.

Question 5

A block, initially at rest, is pulled along a horizontal surface by a force of 10 N making an angle of 60° with the horizontal. There is a constant friction of 2 N between the block and the horizontal surface. What is the gain in kinetic energy of the block after moving 5 m?



Question 5 block and force

A. 33 J

B. 25 J

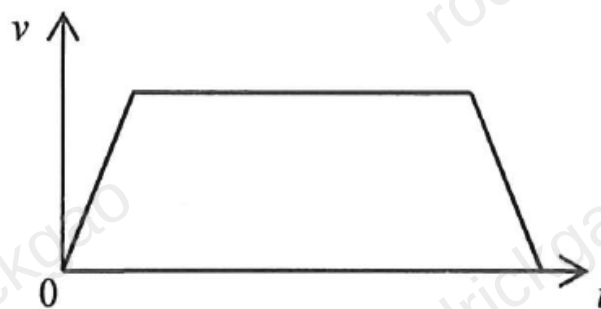
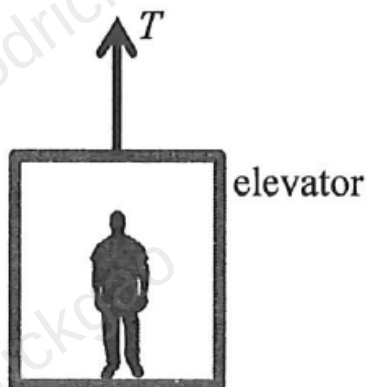
C. 15 J

D. 10 J

Official answer: C

By the work–energy theorem, $\Delta K = W_{\text{net}} = (10 \cos 60^\circ - 2) \times 5 = 15 \text{ J}$. Answer: C.

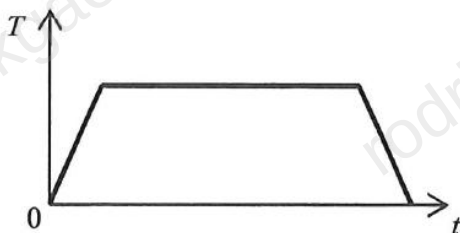
Question 6



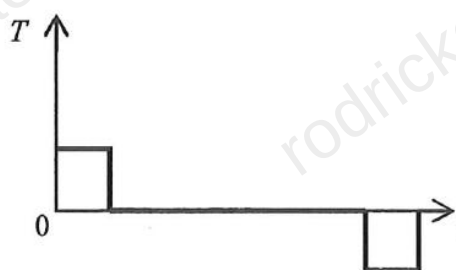
Question 6 elevator and velocity graph

An elevator is pulled upward by a cable. The v - t graph above shows the variation of the elevator's velocity with time. Which of the following correctly indicates the variation of the tension T in the cable with time t ?

A.



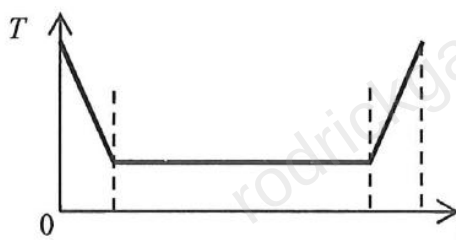
B.



C.



D.



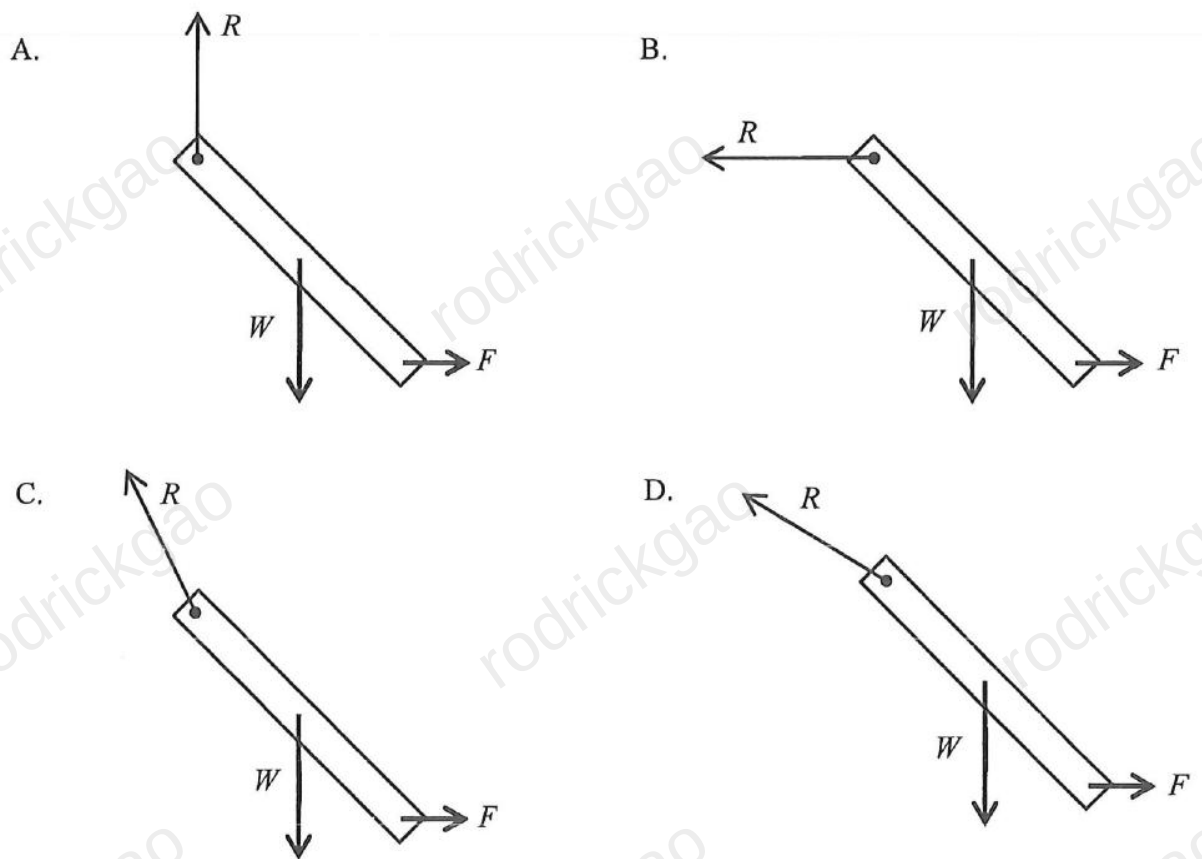
Question 6 graphical options

Official answer: C

From $T - mg = ma$: $T > mg$ while accelerating upward, $T = mg$ at constant speed, and $T < mg$ while slowing down upward. Answer: C.

Question 7

A uniform rod of weight W hinged at its upper end is at rest under a horizontal force F applied at its lower end. Which figure correctly shows the force R acting on the rod at the hinge?



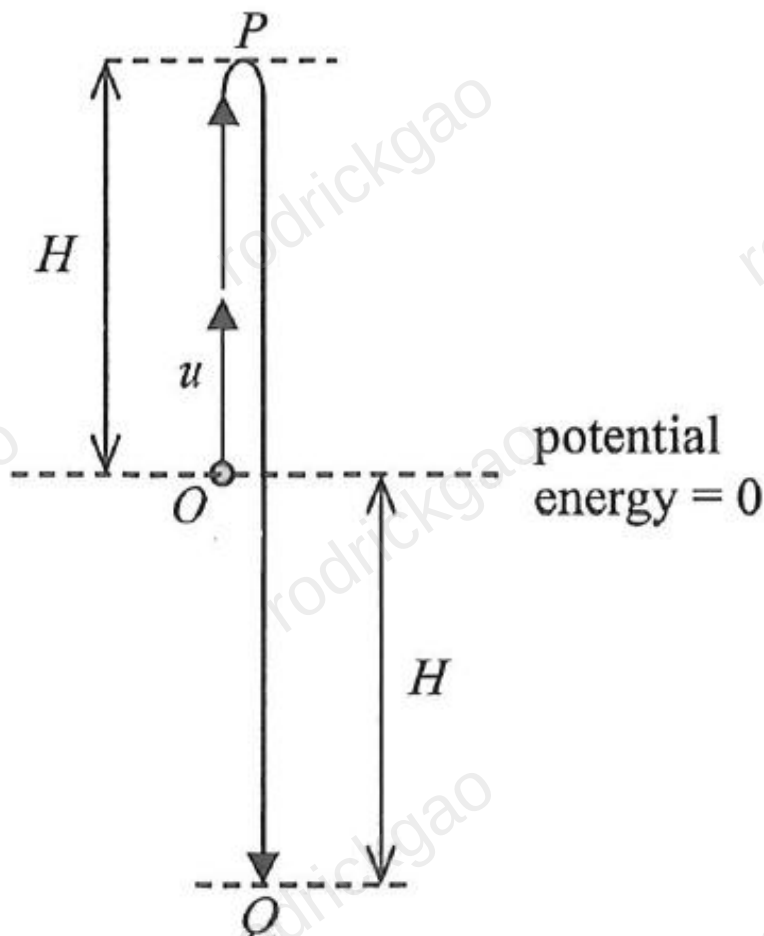
Question 7 graphical options

Official answer: C

Taking moments about the hinge: $F(L/\sqrt{2}) = W[L/(2\sqrt{2})]$, so $F = W/2$. Force balance gives $R_x = -F$ and $R_y = W$; R therefore points steeply upward-left. Answer: C.

Question 8

A small ball of mass m is thrown vertically upward from point O with an initial speed u . It reaches the highest point P and then falls to point Q such that $OP = OQ = H$. The ball's potential energy at point O is taken to be zero.



Question 8 vertical motion

What are the kinetic energy and potential energy of the ball at Q ? Neglect air resistance.

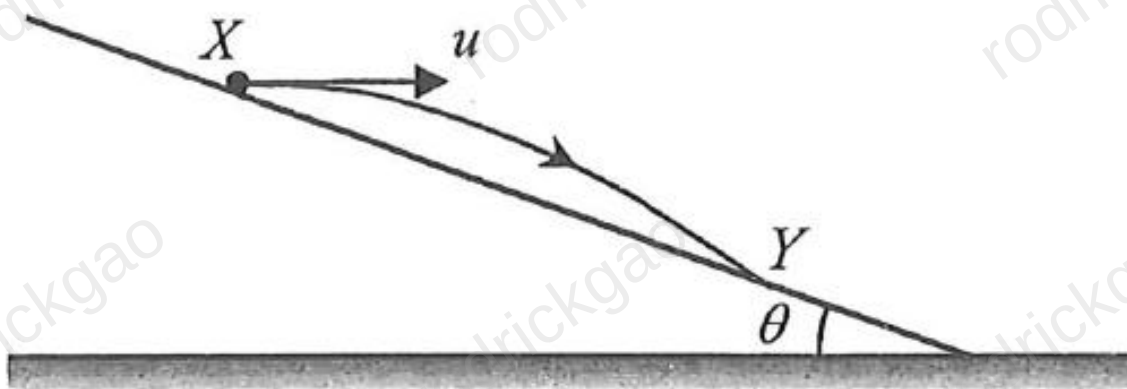
- A. Kinetic energy at Q : $m u^2$; potential energy at Q : $-m g H$
- B. Kinetic energy at Q : $m u^2$; potential energy at Q : $m g H$
- C. Kinetic energy at Q : $\frac{1}{2}m u^2$; potential energy at Q : $-m g H$
- D. Kinetic energy at Q : $\frac{1}{2}m u^2$; potential energy at Q : $m g H$

Official answer: A

At the highest point, $\frac{1}{2}mu^2 = mgH$. At Q , $U = -mgH = -\frac{1}{2}mu^2$; total energy is still $\frac{1}{2}mu^2$, so $K_Q = mu^2$. Answer:

A.

Question 9



Question 9 inclined plane

X and Y are two points on an incline making an angle θ with the horizontal. A particle projected horizontally from X with speed u lands at Y as shown. Which expression correctly gives its time of flight from X to Y? Neglect air resistance.

- A. $(u \sin \theta)/(2g)$
- B. $(u \tan \theta)/(2g)$
- C. $(2u \sin \theta)/g$
- D. $(2u \tan \theta)/g$

Official answer: D

Horizontally $x = ut$ and vertically $y = \frac{1}{2}gt^2$. Since Y lies on the incline, $y = x \tan \theta$; hence $t = 2u \tan \theta/g$. Answer: D.

Question 10

Loosening a screw requires a pair of forces of 10 N acting tangentially as shown in Figure (1). A suitable screwdriver with a handle of diameter 0.025 m shown in Figure (2) is chosen. Estimate the magnitude of the pair of forces needed to act tangentially on the handle of the screwdriver.

Figure (1)

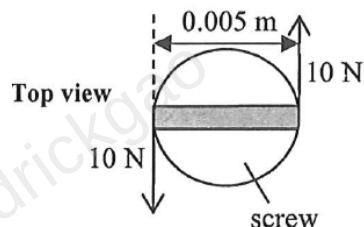
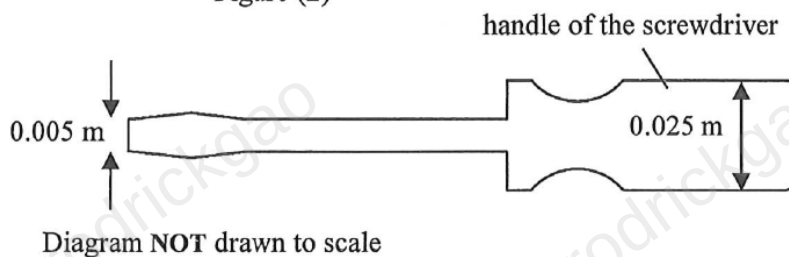


Figure (2)



Question 10 force figures

- A. 2 N
- B. 4 N
- C. 5 N
- D. 10 N

Official answer: A

Equate the moments of the couples: $10 \times 0.005 = F \times 0.025$, giving $F = 2$ N. Answer: A.

Question 11

A car of weight W travels with constant speed on a circular humpback bridge. At the moment when the car passes the top of the bridge, the normal reaction acting on it by the bridge is R . What is the centripetal force acting on the car at that moment?



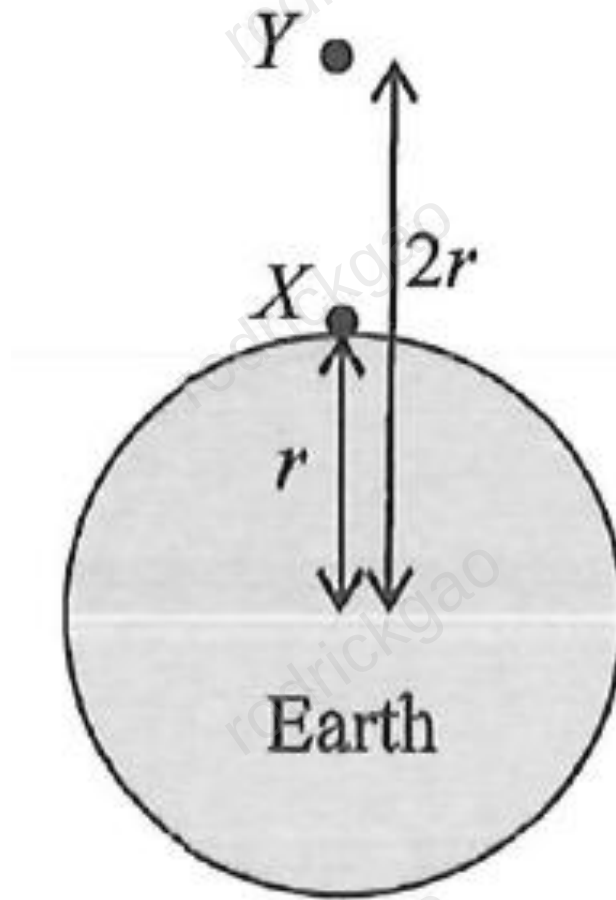
Question 11 hump-backed bridge

- A. $W+R$
- B. $W-R$
- C. W
- D. R

Official answer: B

At the top of the bridge, the inward direction is downward, so $F_c = W-R$. Answer: B.

Question 12

*Question 12 Earth and object*

Two identical objects located at point X and point Y are at distances r and $2r$ from the Earth's centre. The weight of the object at point X is W . What is the weight of the object at point Y?

A. $0.25W$

B. $0.5W$

C. $0.75W$

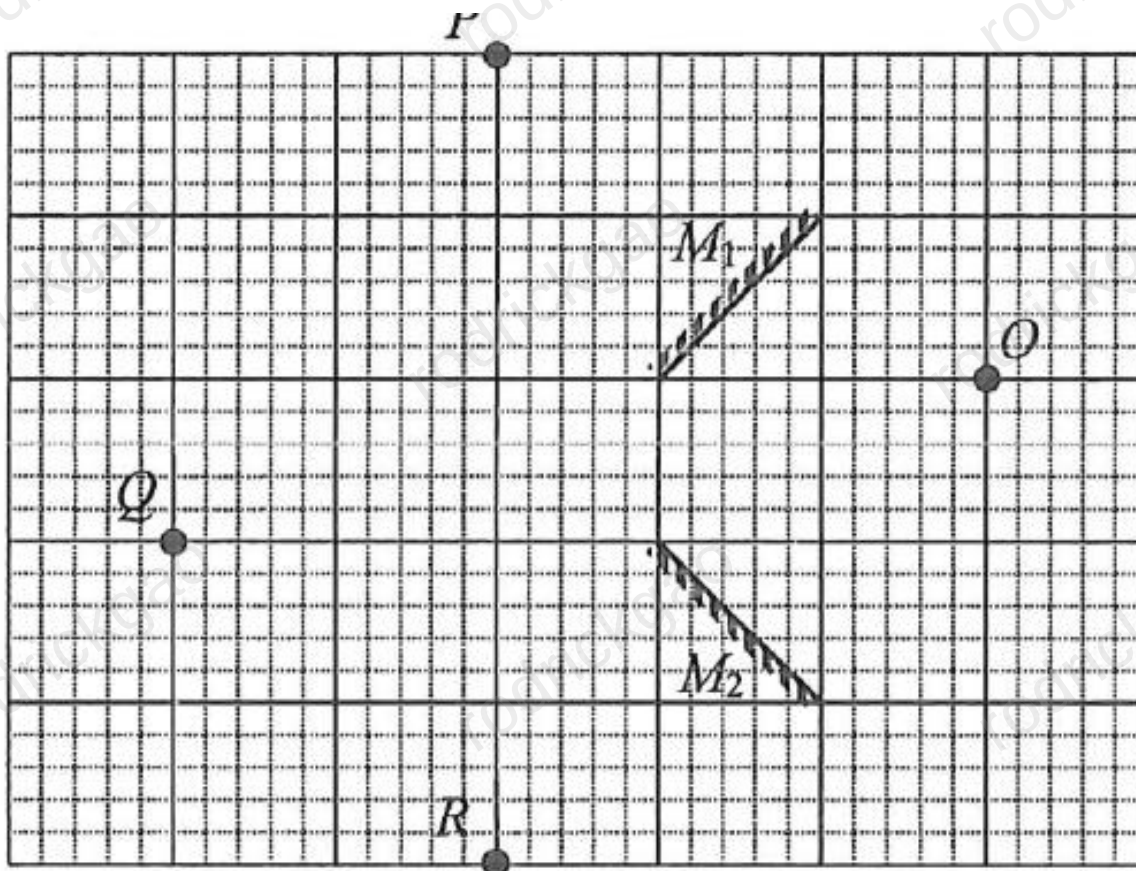
D. W

Official answer: A

Since $W \propto 1/r^2$, doubling the distance changes the weight to $W/4 = 0.25W$. Answer: A.

Question 13

A point object O is placed in front of two plane mirrors M_1 and M_2 as shown.



Question 13 mirror

At which position(s) can an image be formed?

A. P, Q and R

B. Q and R only

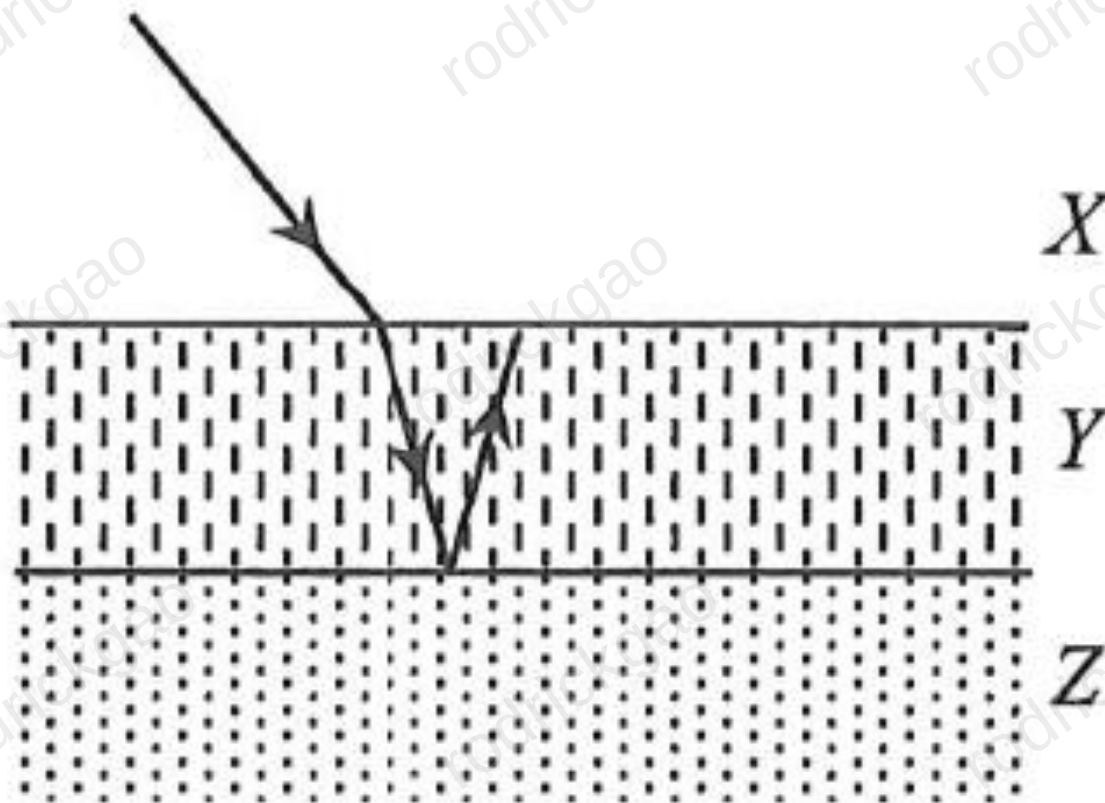
C. R only

D. P only

Official answer: B

A plane-mirror image must be symmetric with the object about the mirror. Reflection in M_1 and M_2 gives Q and R respectively; P is impossible. Answer: B.

Question 14



Question 14 refraction

The figure shows three media X, Y and Z separated by parallel boundaries. A light ray travelling from X into Y then undergoes total internal reflection at the boundary between Y and Z. Arrange in descending order the speed of light in the respective media.

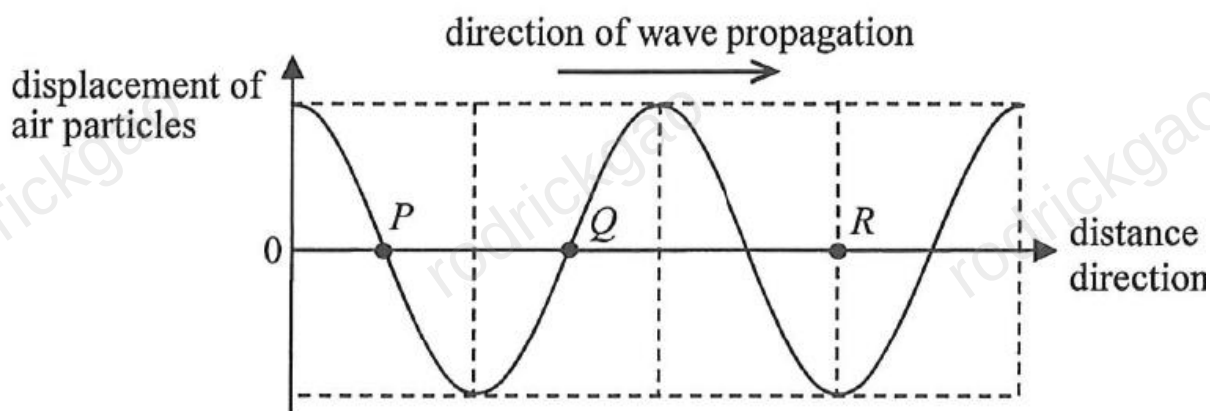
- A. X Y Z
- B. X Z Y
- C. Y Z X
- D. Z X Y

Official answer: D

X→Y bends towards the normal, so $v_X > v_Y$. Total internal reflection at Y→Z gives $v_Z > v_Y$; with parallel boundaries and Snell's law, $v_Z > v_X$. Thus $v_Z > v_X > v_Y$. Answer: D.

Question 15

The graph shows the variation of the displacement of air particles from their equilibrium positions with distance for a sound wave at a certain instant. Displacement along the direction of wave propagation is taken to be positive.



Question 15 displacement graph

Which statement concerning the wave at that instant is correct?

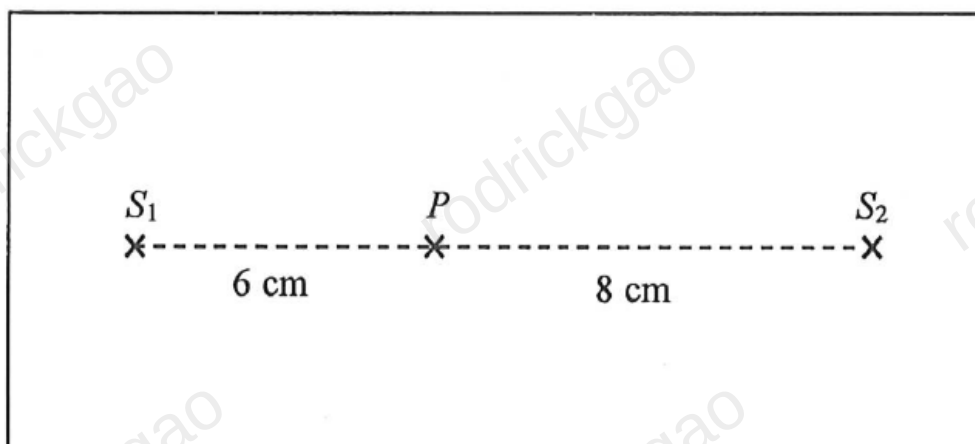
- A. An air particle at P is momentarily at rest.
- B. A rarefaction occurs at Q.
- C. A compression occurs at R.
- D. An air particle at R is moving with the greatest speed.

Official answer: B

For a longitudinal wave, the displacement gradient determines the change in particle separation. The gradient at Q is positive, so neighbouring particles separate and Q is a rarefaction. Answer: B.

Question 16

In the ripple tank shown below, dippers S_1 and S_2 are vibrating in phase and generate water waves of wavelength 0.8 cm towards each other at the same speed. For a particle P at distances 6 cm and 8 cm from S_1 and S_2 respectively as shown, the amplitude of the waves reaching P from S_1 is 1 mm while that from S_2 is 2 mm .



Question 16 ripple tank

What is the amplitude of oscillation of P ?

- A. 0 mm
- B. 1 mm
- C. 2 mm
- D. 3 mm

Official answer: B

The path difference is $8 - 6 = 2.0\text{ cm} = 2.5\lambda$, so the waves arrive in antiphase. Resultant amplitude $= |2 - 1| = 1\text{ mm}$. Answer: B.

Question 17

The light from a sodium lamp consists of two wavelengths 589.00 nm and 589.59 nm . Find the angle between these two spectral lines in the second-order after the light passes through a diffraction grating of 600 lines per mm .

- A. 0.058°
- B. 0.041°
- C. 0.022°
- D. 0.021°

Official answer: A

$d = 1/(600 \times 10^3)\text{ m}$ and $d \sin \theta = m\lambda$. Using $m = 2$ for the two wavelengths gives an angular separation of about 0.058° . Answer: A.

Question 18

In an experiment, monochromatic light is incident normally on a double slit and a diffraction grating placed in turn at the same position. The respective interference and diffraction patterns formed on the screen are observed. Which of the following statements about the patterns is/are correct?

- (1) The fringe separation of the interference pattern is wider.
- (2) The fringes of the diffraction pattern are sharper.
- (3) The width of the whole diffraction pattern is wider.

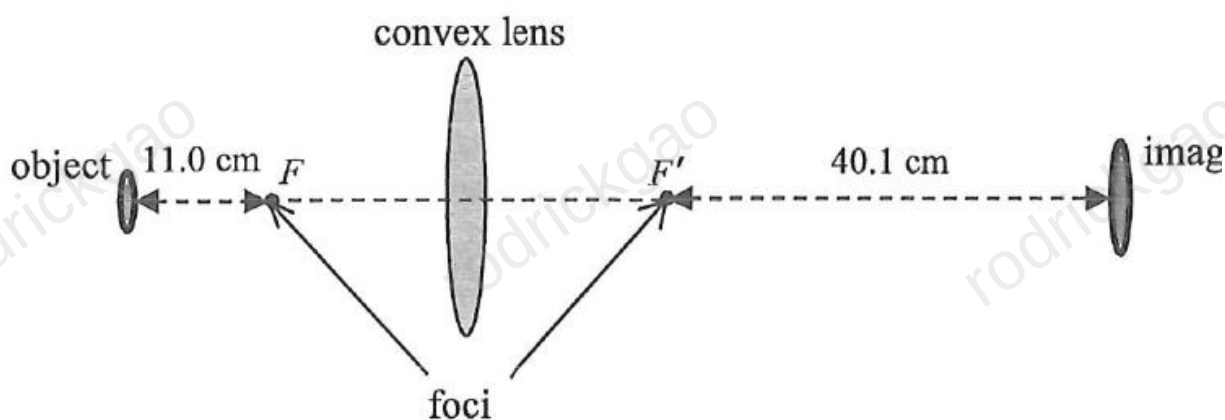
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: D

A diffraction grating has many slits, giving sharper principal maxima; its smaller slit spacing also spreads the spectrum over a wider angle. Statements (2) and (3) are correct. Answer: D.

Question 19

An object is placed at a certain distance from a convex lens; the position of a screen is adjusted until a sharp image is formed.



Question 19 lens

Given that the distances of the object and the image to the respective foci F and F' of the convex lens are 11.0 cm and 40.1 cm as shown, the focal length of the lens is

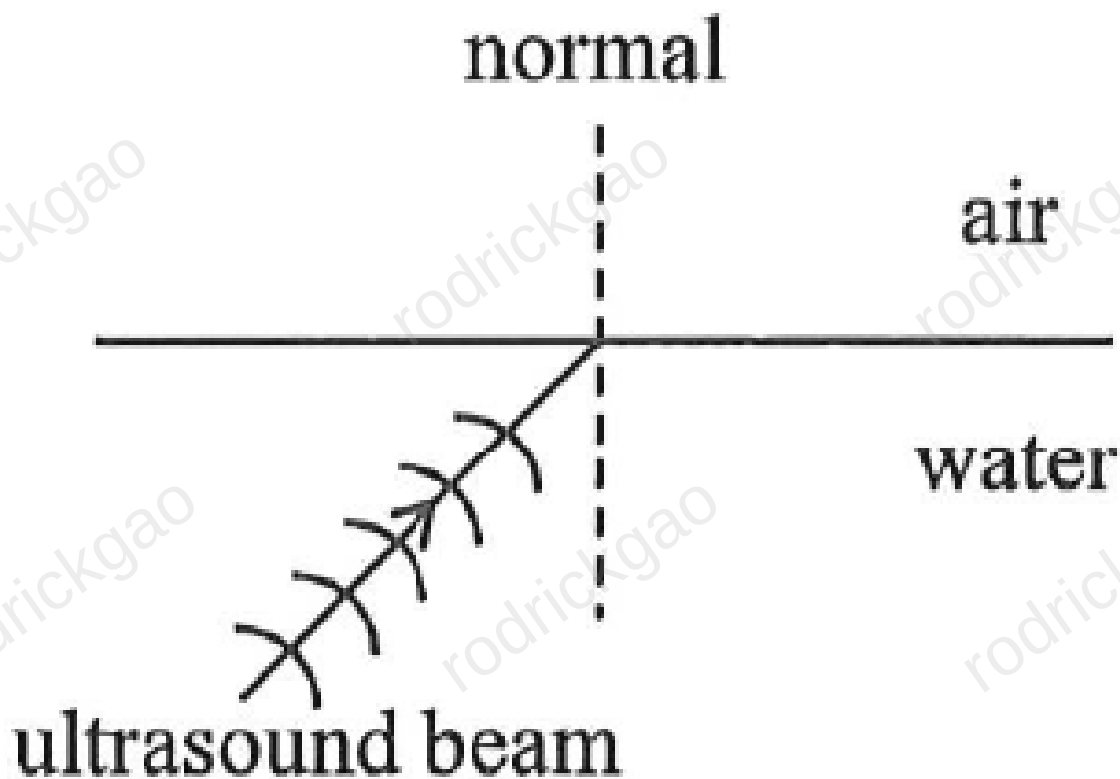
- A. 8.6 cm.
- B. 17.3 cm.
- C. 21.0 cm.
- D. 25.9 cm.

Official answer: C

Using Newton's lens formula $xx' = f^2$: $f = \sqrt{(11.0 \times 40.1)} = 21.0$ cm. Answer: C.

Question 20

A source in water emits a beam of ultrasound towards the water-air interface as shown.



Question 20 ultrasound

Which of the following statements is INCORRECT?

- A. The frequency of the refracted beam is the same as that of the incident beam.
- B. The speed of the refracted beam becomes smaller.
- C. The refracted beam leaving the interface bends towards the normal.
- D. If the angle of incidence is large enough, total internal reflection may occur.

Official answer: D

From water to air, frequency is unchanged, sound speed decreases and the ray bends towards the normal. Total internal reflection is possible only from the slower to the faster sound medium, so statement D is incorrect.

Answer: D.

Question 21

Which of the following phenomena can be explained by the speed of waves being different in different media?

- (1) echoes
- (2) rainbows
- (3) mirages
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

A rainbow and a mirage involve refraction because wave speed changes between media; an echo is reflection. Hence (2) and (3). Answer: C.

Question 22

Three light conducting spheres are suspended by nylon threads. Any two of them are found to attract each other when they are brought close to each other. Which conclusion below is correct?

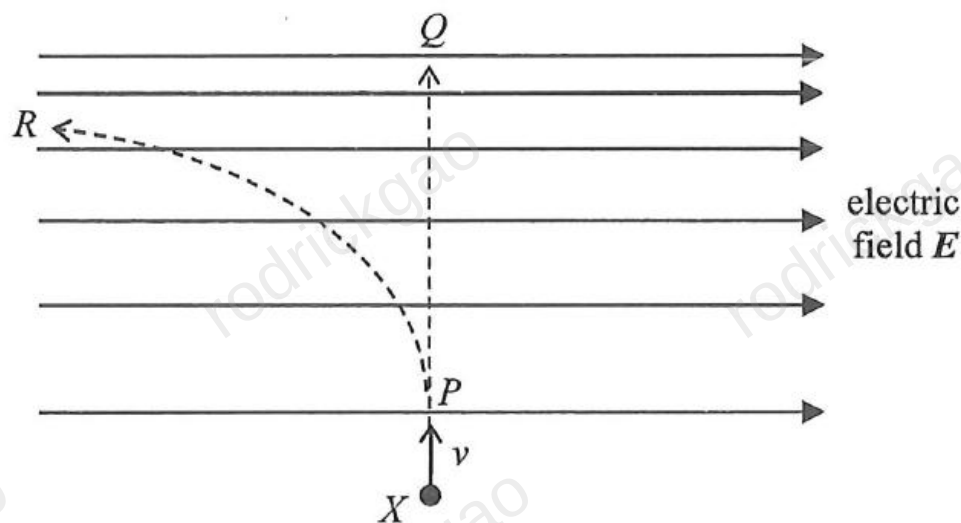
- A. All three spheres are charged.
- B. Only one sphere is charged.
- C. Only one sphere is uncharged and the other two carry like charges.
- D. Only one sphere is uncharged and the other two carry unlike charges.

Official answer: D

A neutral conductor can attract either charged sphere by induction. The other two spheres must carry opposite charges to attract each other. Answer: D.

Question 23

A negatively charged particle X enters an electric field with initial velocity v perpendicular to the field lines as shown. Neglect the effects of gravity.

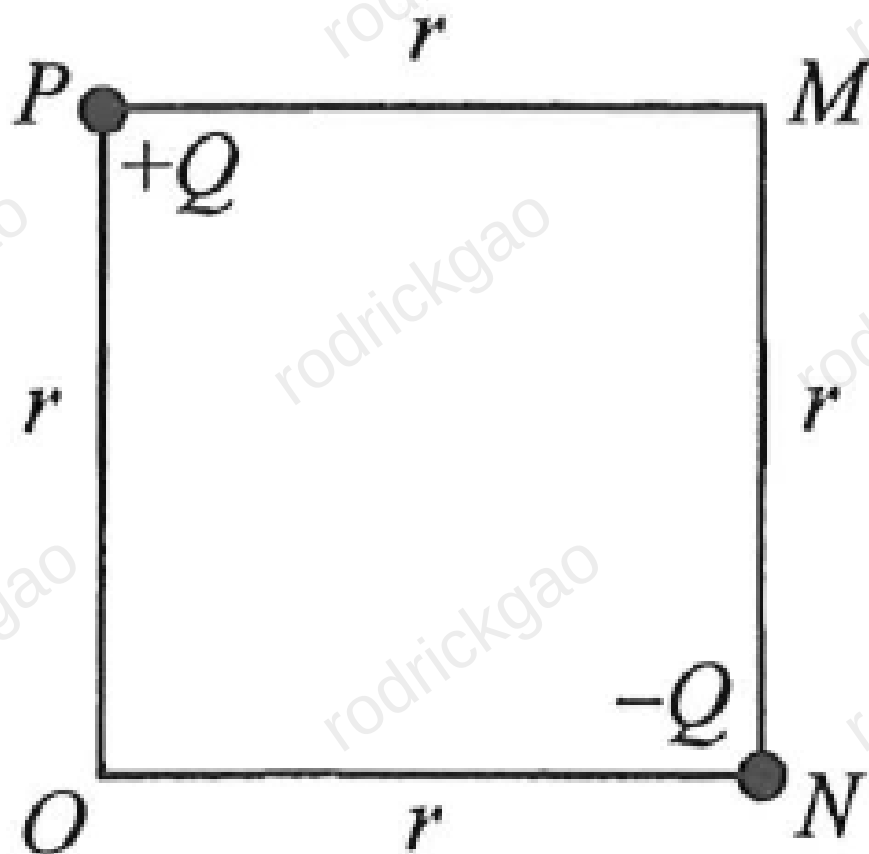
Top view **X may move along***Question 23 electric field*

- A. PQ with uniform speed.
- B. PQ with increasing speed.
- C. PR with decreasing speed.
- D. PR with increasing speed.

Official answer: D

A negative charge experiences force opposite to the electric field, so its path bends along PR. The electric force then has a component along its velocity and does positive work, increasing its speed. Answer: D.

Question 24



Question 24 charged square

Point charges $+Q$ and $-Q$ are fixed respectively at corners P and N of square MNOP of side length r as shown above. What is the electric field at O?

- A. $[\sqrt{2} Q/(4\pi \epsilon_0 r^2)]$ along the direction from P to N
- B. $[Q/(4\pi \epsilon_0 r^2)]$ along the direction from P to N
- C. $[\sqrt{2} Q/(4\pi \epsilon_0 r^2)]$ along the direction from N to P
- D. $[Q/(4\pi \epsilon_0 r^2)]$ along the direction from N to P

Official answer: A

At O, the fields due to $+Q$ and $-Q$ each have magnitude kQ/r^2 and are perpendicular. Their resultant is $\sqrt{2}kQ/r^2$, directed from P to N. Answer: A.

Question 25

A student tests an electric circuit by connecting a filament light bulb to a dry cell using two connecting wires. The bulb does not light up. Which of the following is certainly NOT a possible explanation?

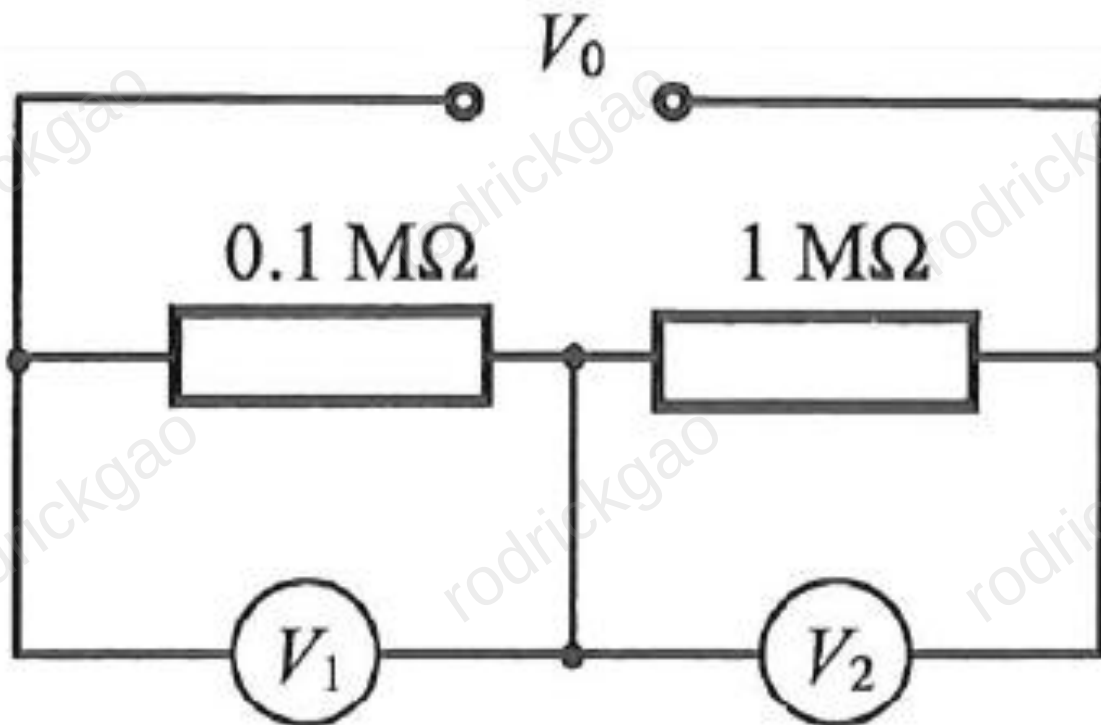
- A. The dry cell may be flat.
- B. The circuit may be open due to a bad connection.
- C. The dry cell may have its polarities connected the wrong way round.
- D. The bulb may have blown.

Official answer: C

A filament lamp has no polarity, so reversing the cell still permits current. Each other change can make the lamp fail to light. Answer: C.

Question 26

In the circuit below, a voltage V_0 is applied across resistors of $0.1\text{ M}\Omega$ and $1\text{ M}\Omega$ connected in series. Voltmeters V_1 and V_2 are connected respectively across the two resistors as shown. V_1 is ideal while V_2 has an internal resistance of $1\text{ M}\Omega$.



Question 26 circuit

Estimate the ratio of the voltmeter readings $V_1 : V_2$.

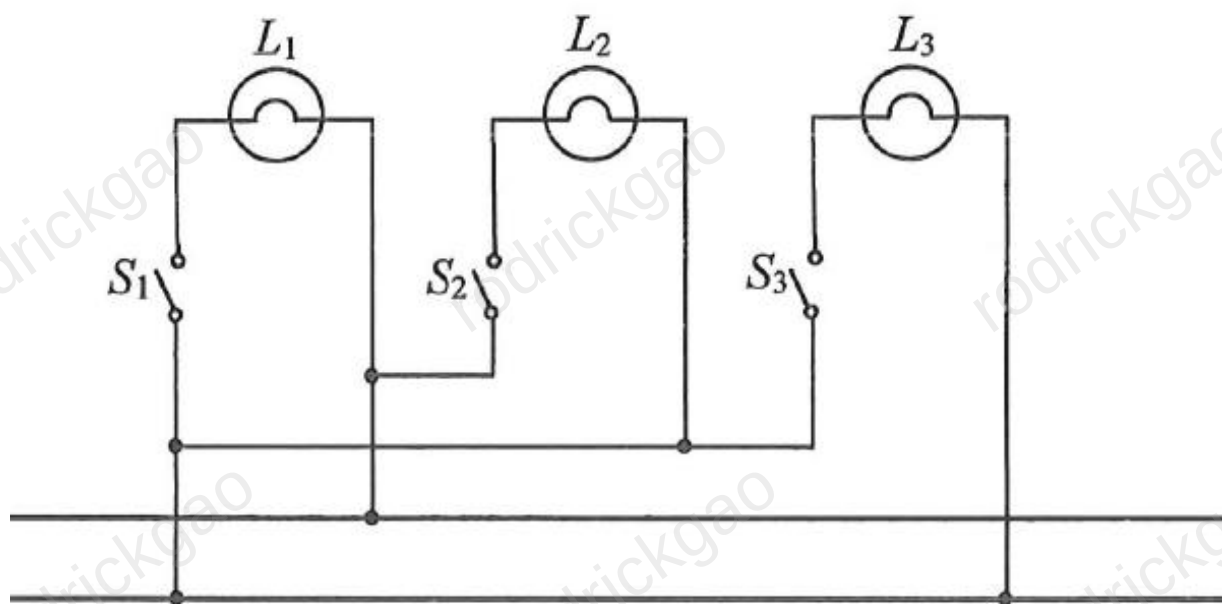
- A. 5:1
- B. 1:5
- C. 10:1
- D. 1:10

Official answer: B

For V_2 , the $1\text{ M}\Omega$ internal resistance is parallel to the $1\text{ M}\Omega$ resistor, giving $0.5\text{ M}\Omega$. The series divider gives $V_1 : V_2 = 0.1 : 0.5 = 1 : 5$. Answer: B.

Question 27

The figure shows how lamps L_1 , L_2 and L_3 are connected in the ring circuit of a house. There are some wrong connections in the circuit.



Question 27 household circuit

Live Neutral

If switches S_2 and S_3 are closed while switch S_1 remains open, which of the lamps will light up?

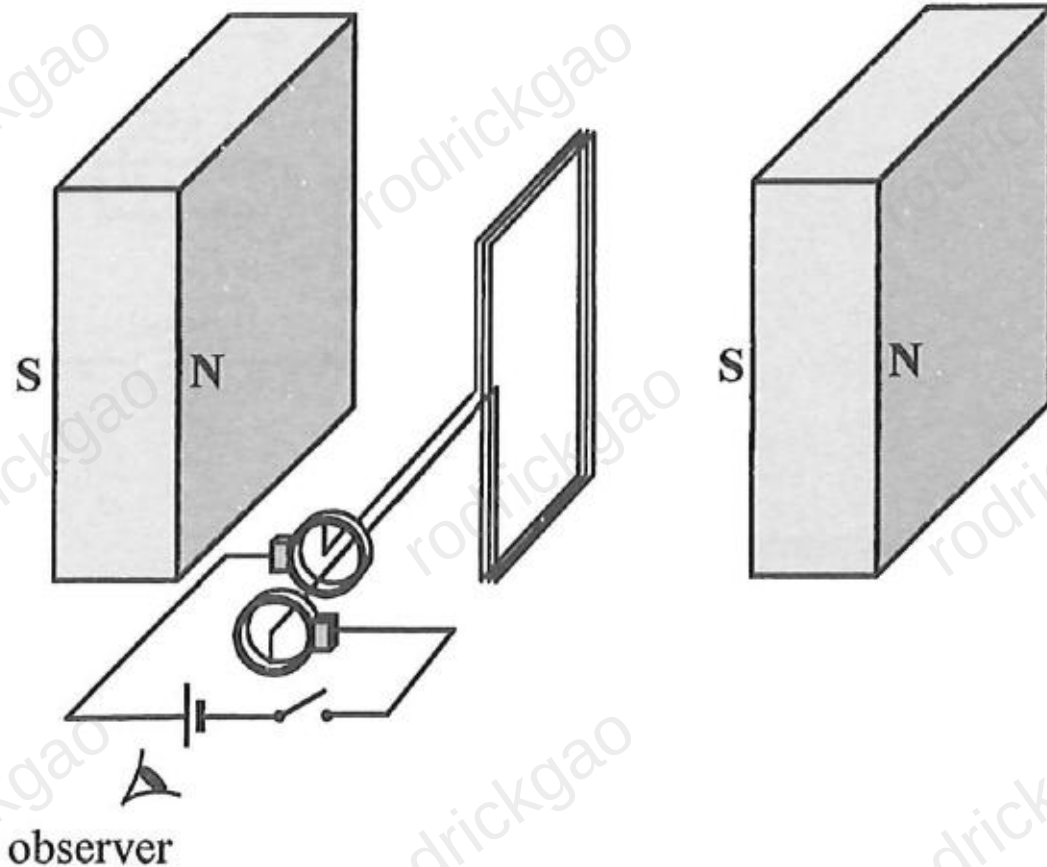
- A. L_2 only
- B. L_3 only
- C. L_2 and L_3 only
- D. none of them

Official answer: A

Tracing the wiring after S_2 is closed, L_2 is connected between live and neutral and forms a complete circuit; L_3 has no required potential difference. Only L_2 lights. Answer: A.

Question 28

The figure shows a rectangular coil connected in series with a dry cell and a switch through a pair of slip rings. The magnetic field between the two magnets is uniform and is perpendicular to the plane of the coil at the moment shown.



Question 28 coil and magnets

When the switch is closed, the coil will

- A. remain at rest.
- B. rotate in a clockwise direction continuously.
- C. rotate in an anticlockwise direction continuously.
- D. change the direction of rotation after turning through 180° .

Official answer: A

The field is perpendicular to the coil, so the coil's normal is parallel to $\mathbf{B}$. Hence $\tau = NIAB \sin \theta = 0$ and the coil remains stationary. Answer: A.

Question 29

A student uses a search coil to measure the magnetic field of a permanent magnet. However, no reading is observed on the CRO connected to the search coil. Which of the following is the possible reason?

- A. The magnet is not strong enough.
- B. The magnetic field produced by the magnet is constant.**
- C. The sensitivity of the CRO is too low.
- D. The size of the search coil is not large enough to pick up sufficient magnetic flux.

Official answer: B

A search coil produces an induced e.m.f. only when magnetic flux changes. A stationary permanent magnet gives constant flux, so the CRO shows no signal. Answer: B.

Question 30

An electric iron works at its rated power of 1100 W under a 50 Hz a.c. mains of root-mean-square voltage 220 V. What power will it deliver when working under a 100 Hz a.c. of root-mean-square voltage 110 V?

- A. 275 W**
- B. 778 W
- C. 550 W
- D. 1100 W

Official answer: A

Treating the iron as resistive, $R = 220^2/1100 = 44 \, \Omega$. Therefore $P = 110^2/44 = 275 \, \text{W}$, independent of frequency. Answer: A.

Question 31

A radioactive source undergoes decay such that 300 nuclides have decayed in 1 minute. Its activity is

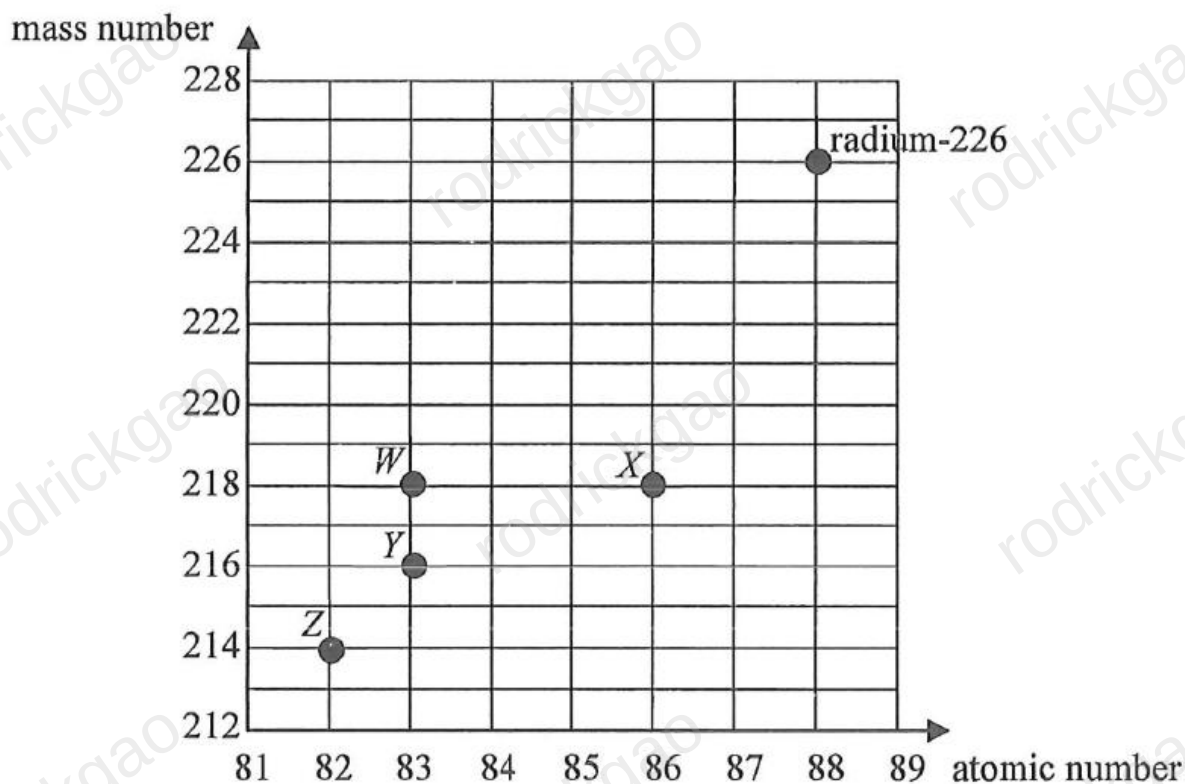
- A. 0.2 Bq.
- B. 5 Bq.**
- C. 300 Bq.
- D. 18000 Bq.

Official answer: B

Activity $A = \text{number of decays/time} = 300/60 = 5 \, \text{s}^{-1} = 5 \, \text{Bq}$. Answer: B.

Question 32

Radium-226 undergoes a series of α and β decays before finally becomes a stable nuclide. Referring to the plot of mass number against atomic number below, which nuclides, W, X, Y and Z, cannot be possible members in the decay series of radium-226?



Question 32 decay graphs

A. W and X

B. W and Y

C. Z and X

D. Z and Y

Official answer: B

An α decay changes $(A, Z) \rightarrow (A-4, Z-2)$, while β^- changes $(A, Z) \rightarrow (A, Z+1)$. Starting from $(226, 88)$, A must be $226-4n$: Y has impossible $A = 216$, and at $A = 218$ the atomic number cannot fall to 83. W and Y are impossible. Answer: B.

Question 33

In the following nuclear reaction, nuclide X and Y fuse to form nuclide Z under suitable conditions and the amount of energy released is 7.16 MeV.



The total mass of X and Y is

- A. 0.00385 u less than the mass of Z.
- B. 0.00385 u more than the mass of Z.
- C. 0.00769 u less than the mass of Z.
- D. 0.00769 u more than the mass of Z.

Official answer: D

Energy release means the reactants are heavier. $\Delta m = 7.16/931.5 = 0.00769$ u, so the total mass of X and Y exceeds that of Z by 0.00769 u. Answer: D.